

Hayden Choi

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Education

McMaster University

Honours Computer Science Co-op, Bachelor of Applied Science (GPA: 3.9/4.0)

Expected May 2027

Hamilton, ON

Experience

Software Engineer Intern

May 2026 – Present

Trend Micro

Ottawa, ON

- Developed a data analysis framework over **1.2M+** customer risk events using **AWS (Athena, Glue, S3)** and **Apache Iceberg** to identify the most-remediated risks across **39** tenants and flag low-value risk events
- Developed a **KNN-based telemetry classifier** with **statistical distance weighting** on Trend Vision One's **XDR platform** to automate **multi-class threat categorization** correctly identifying **230+** unique threats
- Automated Trend Vision One policy updates via **GitHub Actions**, cutting **4+** hours of manual work per sprint

Software Engineer Intern

Dec 2025 – Apr 2026

Knockri

Toronto, ON

- Architected an end-to-end **ReAct AI** agent on IBM Watsonx Orchestrate that autonomously resolves customer support tickets, reducing ticket volume by **83%** and external support costs by **60%**
- Reduced transcription tool latency **74%** by resolving **N+1 GraphQL** queries and refactoring data-fetching architecture
- Engineered **Knockri CLEAR**, a secure proctoring suite using Django and React that monitors and validates **10k+** assessment sessions across Fortune 500 and government clients
- Implemented a **Heartbeat API** via **GraphQL/UUID4** to restrict **7,000+** duplicate launches via backend locking
- Safeguarded production stability by writing **Cypress E2E** and **Django mock tests** for automated **CI/CD** regression

Software Engineer Intern

May 2025 – Aug 2025

Brighter Signals BV

Markham, ON

- Built a real-time signal processing pipeline for an **F1 plank wear detection** prototype, applying baseline calibration at ingestion to denoise raw sensor data and improve detection accuracy by **64%**
- Developed **5+** specialized GUIs for investor demos that secured **\$1.6M+** in funding for **MVP** development
- Optimized **4-axis Control Logic** by prioritizing event-driven signal polling to reduce real-time gesture latency by **35%**
- Shifted **Signal Pre-processing** to embedded firmware to automate filtering, reducing frontend compute load by **76%**
- Co-authored an **IROS 2025** paper and named a **Best Poster Finalist** for research on data-rich tactile sensors

Software Engineer Intern

May 2024 – Sept 2024

Mana Immigration

Markham, ON

- Architected **6+** full-stack Flutter apps for international clients using a Factory Pattern for reusable UI components
- Boosted app responsiveness by **50%** by implementing **Deferred Loading** for complex navigation modules
- Engineered a **Discovery Engine** via **Spotify REST APIs** that bypassed algorithmic bias for genre-isolated playlists
- Migrated legacy **WordPress** infrastructure to **React.js**, refactoring monolithic pages into reusable components

Projects

ColdApproach AI | *Next.js, FastAPI, AWS (SES, SQS, Lambda), OpenAI, Hunter.io*

GitHub | Website

- Built an AI outreach platform with **FastAPI** and **Next.js** using **Hunter.io** discovery and a two-pass **GPT** pipeline
- Engineered an async batch pipeline with **AWS SQS** to process up to **20** URLs through scraping and generation stages
- Implemented **EventBridge Scheduler** with **AWS Lambda** for delayed email delivery via a provider factory pattern
- Designed a **Next.js** and **TypeScript** frontend with **Zustand** persistence and **React Query** polling for batch tracking

SyllaScan | *React, Python, OpenAI, AWS, FastAPI, Google Calendar API*

GitHub

- Developed an **NLP pipeline** using **FastAPI** to parse unstructured PDF syllabi into validated **JSON** schemas
- Applied **Prompt Engineering** strategies to extract complex date-event pairs with high accuracy via **GPT-4**
- Integrated **Google OAuth 2.0** and **Calendar API** to automate bulk event synchronization for academic schedules
- Built a responsive **React** interface for real-time file uploads and interactive schedule previews before final sync

Technical Skills

Languages: Python, JavaScript, Java, TypeScript, Ruby, C++, C, SQL

Libraries/Frameworks: Django, React, PyTorch, Scikit-learn, FastAPI, PySpark

Databases/Cloud: AWS (S3, Glue, Athena, SES, SQS), Azure, Apache Iceberg, MongoDB

Tools/Methodologies: Git, Docker, Linux, GraphQL, Cypress, CI/CD, Agile